Channel Estimation for Rydberg Receivers Using Learned Hankel Filtering

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***Abstract*—Rydberg receivers use excited atoms to sense radio signals. Magnitude-only observations do not directly reveal the wireless channel. Channel structure can help recover it, but fixed filtering may remove useful details. We study a small learned estimator that alternates measurement fitting with Hankel filtering. It learns from withheld observations, without true-channel targets. Simulations show lower average error and processing time than the tested classical Hankel estimator. Both structure and noise information help average accuracy. However, the classical method remains better with a weak reference and in some few-path channels. Learned filtering is a useful option for the tested model within these operating limits.**

***Index Terms*—Channel estimation, Hankel filtering, magnitude measurements, Rydberg receivers.**

I. Introduction

RYDBERG atoms respond to radio-frequency electric fields, offering a way to study new wireless receivers. Here, signals from several users combine with a known reference. The receiver measures their combined magnitude, not their full complex value. The channel describes changes in signal strength and phase on the way to the receiver. Model-based methods and learned methods such as URformer estimate it [1], [2].

A few propagation paths can create a repeated pattern across a regular array. Arranging channel samples in a Hankel matrix lets an estimator use this pattern. Strong filtering can remove real details; weak filtering can leave noise. Searching many filtering settings also takes time. We study a small model that learns how much filtering to apply. The true channel is never shown to it during training. Instead, some measurements are hidden from the estimator and used to check its predictions. This is called self-supervised learning.

Our contributions are: (i) a 188-parameter estimator that combines Rician measurement updates with smooth Hankel filtering; (ii) training this estimator on withheld measurements, without channel targets or channel-error checkpoint selection; and (iii) paired tests of accuracy, processing time and simpler versions, including fresh-test replication and input-removal tests. We report weak-reference and few-path failures alongside the gains.

II. Related Work

Training on withheld measurements is established in image reconstruction [3] and self-supervised denoising is used for Rydberg sensor signals [4]. Huang et al. learn Hankel-factor updates from fully sampled targets [5]. Luo et al. combine self-supervision and structured low-rank reconstruction for complex MRI data [6]. DeepTensor fits network-based low-rank factors to an observed object [7]. Rician-likelihood training for magnitude data is also established [8]. Neither self-supervision, learned Hankel processing nor the Rician loss is new on its own.

We recover a complex multiuser channel from noisy magnitudes and a known reference, rather than restoring measured complex samples. A small controller sets filtering strength instead of generating low-rank factors. Our companion manuscript studies classical Hankel estimation and rank selection [9]; here we learn filtering without channel labels.

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III. Receiver and Channel Model

An array with N elements receives P known training symbols, or pilots, from each of K users. The observations are

$$Z = |GS + B + W|, \qquad W_{np} \sim \mathcal{CN}(0, v). \tag{1}$$

Here $G \in \mathbb{C}^{N\times K}$ is the channel, $S \in \mathbb{C}^{K\times P}$ contains the pilots, and $B \in \mathbb{C}^{N\times P}$ is the known reference. Magnitudes act entrywise. Independent noise has total complex variance v, or $v/2$ in each real component. The reference makes magnitudes depend on relative signal phases, but does not ensure accurate recovery in every case. Reference calibration and noise variance are known, except in stated noise-input tests. Receiver conversion gain is absorbed into the scaling.

For the half-wavelength uniform linear array,

$$g_{nk} = \sum_{\ell=1}^{L_k} \alpha_{\ell k} e^{-j\pi n \sin\theta_{\ell k}}, \quad n = 0, \ldots, N-1. \tag{2}$$

Independent gains follow $\mathcal{CN}(0, 1/L_k)$, giving each user unit average channel power. Angles are uniform on $[-\pi/2, \pi/2]$; pilot entries follow $\mathcal{CN}(0, 1)$. The reference has a known spatial response and constant waveform. Define $\mathrm{SNR} = 10\log_{10}(K/v)$ and $\mathrm{RSR} = 10\log_{10}|\alpha_b|^2$, where α_b is reference amplitude. RSR uses one user's power: for three users, RSR 12 dB is about 7.23 dB relative to their combined power.

IV. Proposed Estimator

The estimator starts with a rough channel guess. It repeats two steps ten times: one pulls the guess toward the measurements, and the other smooths it using the array structure. A small network decides how much smoothing to apply at each step (Fig. 1).

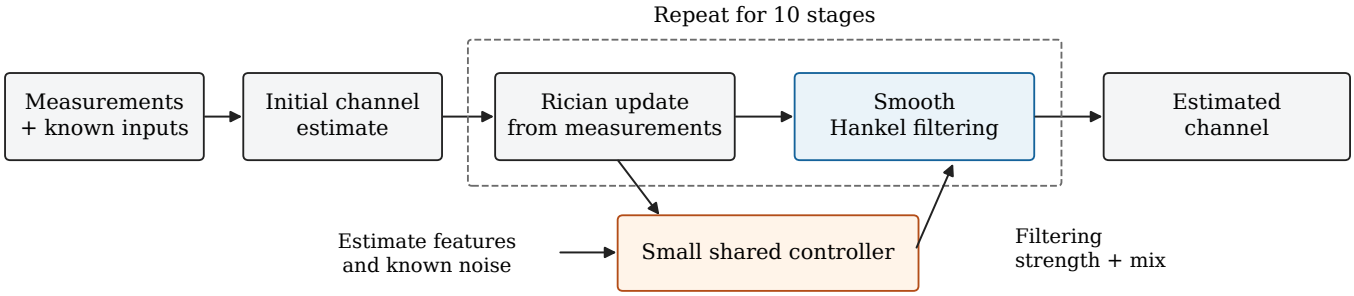


Fig. 1. Estimator overview. The controller chooses filtering strength and mixing. During training, nine withheld pilots score predictions but do not enter initialization or the ten stages.

A. Measurement Update

The magnitude of a complex signal with Gaussian noise follows a Rician distribution. For noiseless magnitude $a = |\mu|$, its density is

$$p(z \mid a, v) = \frac{2z}{v} e^{-(z^2+a^2)/v} I_0(2za/v), \tag{3}$$

where I_0 is a modified Bessel function. Let $M = GS + B$ and $\kappa = 2Z|M|/v$. Entrywise, form

$$Q = Z \frac{I_1(\kappa)}{I_0(\kappa)} \frac{M}{|M|}, \tag{4}$$

using the continuous limit at $M = 0$. The expectation-maximization (EM) update and learned step are

$$G_{\mathrm{EM}} = (Q - B)S^H(SS^H)^{-1}, \tag{5}$$

$$G_{\mathrm{dc}} = G + \gamma_t(G_{\mathrm{EM}} - G), \quad \gamma_t = 2\,\mathrm{sigmoid}(a_t). \tag{6}$$

A linear solve evaluates the inverse. Input pilots must have full row rank. Requiring at least $2K$ input pilots is a numerical rule, not a proof of unique recovery.

B. Smooth Hankel Filtering

Each updated user channel forms a Hankel matrix $H = \mathcal{H}(g_{\mathrm{dc},k})$, with equal entries along each antidiagonal. At $N = 32$, it is 17×16. Singular values describe the strengths of its components. As an illustration, a hard rank-two rule changes $[10, 6, 0.3]$ to $[10, 6, 0]$: it deletes the last component. Our smooth rule is

$$\widehat{\sigma}_i = \frac{\sigma_i^3}{\sigma_i^2 + \tau^2}. \tag{7}$$

For $\tau = 1$, the same example becomes approximately $[9.901, 5.838, 0.0248]$. Small components shrink more, without being deleted outright. Some small components may contain real channel details.

Set $q = \lceil N/2 \rceil$, $s_H = \|H\|_F/\sqrt{q}$, and $\tau = s_H[0.001 + \mathrm{softplus}(u)]$. The equivalent operation is

$$\widehat{H} = H - \tau^2 H(H^H H + \tau^2 I)^{-1}, \tag{8}$$

$$g_k^{t+1} = g_{\mathrm{dc},k} + \eta_{t,k}(\mathcal{H}^\dagger \widehat{H}_k - g_{\mathrm{dc},k}), \tag{9}$$

where $\mathcal{H}^\dagger$ averages antidiagonals and $\eta_{t,k} = \mathrm{sigmoid}(w)$. The code normalizes by s_H before solving and guards zero scales. Gradients pass through the solve, without an identity-backward approximation. Neither control estimates path count.

C. Controller Inputs

The network has eight inputs, 16 hidden units, two outputs and SiLU activation $x\,\mathrm{sigmoid}(x)$. Its 178 parameters and ten step sizes give 188. For $p_i = \sigma_i / \sum_j \sigma_j$, its first three inputs are

$$e^{-\sum_i p_i \log p_i}/q, \qquad p_1, \qquad \sum_{i=q-\lceil q/2 \rceil+1}^{q} p_i. \tag{10}$$

They tell the network how spread out the components are, how much the largest contributes, and how much the smaller components contribute. Two more inputs measure disagreement with observations and expected noise relative to estimated channel power:

$$f_4 = \log\left(1 + \frac{\mathrm{mean}[(Z - |G_{\mathrm{dc}}S + B|)^2]}{v}\right), \tag{11}$$

$$f_{5,k} = \log\left(1 + \frac{v\,\mathrm{tr}[(SS^H)^{-1}]}{K\widehat{p}_k}\right), \quad \widehat{p}_k = \|g_{\mathrm{dc},k}\|^2/N. \tag{12}$$

Inputs $P/(2K)$, $\log q$ and $(t+1)/10$ give measurement count, size and stage ($t = 0, \ldots, 9$). Denominator guards avoid division by zero. True channels, path counts and family labels are not supplied. Weights stay fixed.

D. Training Without True Channels

Of 30 pilot columns, 21 enter the estimator and nine are withheld. Training scores predicted withheld magnitudes using the Rician loss

$$\ell(z, \mu, v) = \frac{(z - |\mu|)^2}{v} - \log I_{0e}(2z|\mu|/v). \tag{13}$$

Here I_{0e} is the scaled Bessel function. We omit an observation-only term and use small-field series at continuous limits. The loss fits Rician amplitude, not its mean. Splits change each epoch; separate validation cases with a fixed split select the checkpoint. Deployment uses all pilots, without extra measurements or weight training.

V. Simulation Study

A. Setup and Comparators

Table I gives the settings. Half the training cases have three to seven paths per user; the rest use fixed counts of 2, 4, 8, 12

TABLE I
MAIN SIMULATION AND TRAINING SETTINGS

Setting	Value
Array elements / users / pilots	$32/3/30$
Training / validation cases	16,384/2,048
SNR values (dB)	$-10, -5, 0, 5, 10, 15, 20$
Training RSR values (dB)	$5, 12, 20$
Training seeds	3, sharing training cases
Estimator stages / parameters	$10/188$
Training epochs / batch size	$20/32$
Optimizer / learning rate	Adam / 0.001
Input / withheld pilots	$21/9$
Training / evaluation precision	Single / double

and 16. Validation weights 126 SNR, RSR and path-setting combinations equally. These condition labels set validation weights only; model selection uses withheld likelihood, not channel error.

The main study trains 18 models: two updates, three filtering choices and three seeds. Seeds change weights, batch order and pilot splits, but share training cases. Training took 3.67 hours with overlapping workers, excluding data preparation and preflight. Tests cover 32 conditions with 400 paired cases each. Two noise-input tests reuse cases, giving 12,000 distinct cases. All methods share channels, pilots and noise per case.

HS-GS, the companion study's classical Hankel estimator [9], fits ranks $1, \ldots, q$ on 70% of pilots, using 25 EM updates and four rank/Hankel projection sweeps per update. Squared magnitude residuals on the remaining pilots select a rank. It then refits all pilots for 100 updates. Rank search uses no extra observations and counts toward time. Its rank-selection loss differs from our Rician training loss. Other controls use EM, Rician or amplitude-gradient updates. Global filtering has 30 parameters and scales its threshold with s_H; no filtering has ten. Hard-Hankel Rician uses development-selected rank ten and one projection sweep per update. Constant-output controls replace adaptive outputs with validation averages without retraining.

Our earlier URformer-style model has 1,586,900 parameters and used 80,000 labelled examples, fixed pilots, $P = 20$ and RSR 10 dB. It has no matched evaluation under this paper's protocol; its earlier scores are not included in Table II.

Channel error is normalized mean squared error (NMSE):

$$\mathrm{NMSE}_{\mathrm{dB}} = 10 \log_{10} \frac{\sum_i \|\widehat{G}_i - G_i\|_F^2}{\sum_i \|G_i\|_F^2}. \tag{14}$$

Lower is better. Per-seed dB scores and the seven main SNR settings are averaged equally. Positive gain means lower error. A severe error has more energy than the true channel (per-case linear NMSE above one). Pointwise 95% intervals use 4,000 paired whole-case resamples, conditional on saved models, excluding training-dataset uncertainty. Classical stopping methods have a 1,000-update cap; reaching it need not mean convergence.

B. Accuracy and Processing Time

HS-GS is the strongest tested classical method on the declared average (Table II). The adaptive gradient-descent (GD) core has lower test NMSE than the primary EM core. We retain EM because validation likelihood selected it before test generation.

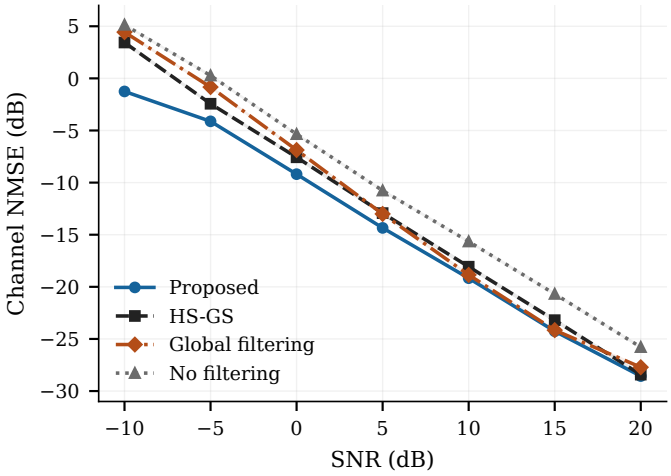


Fig. 2. Main SNR test: $N = 32$, $K = 3$, $P = 30$, RSR 12 dB, and three-to-seven-path channels. Lines average per-seed scores; bands are conditional paired-case 95% intervals.

TABLE II
ALL MAIN-STUDY COMPARATORS

Method	NMSE (dB)	Time (ms)	Severe (%)
EM adaptive (primary)	-14.42	8.24	0.45
EM learned steps only	-10.41	2.98	22.11
EM global filtering	-12.43	5.61	19.19
EM constant-output control	-9.19	5.61	11.42
GD adaptive (secondary)	-14.59	8.00	0.85
GD learned steps only	-10.44	2.73	21.15
GD global filtering	-12.57	5.38	18.37
GD constant-output control	-9.95	5.36	12.36
HS-GS, rank search	-12.74	525.62	14.25
EM, 10 updates	-10.23	4.10	20.64
EM, 100 updates	-10.28	17.01	23.57
Rician GD, 10 updates	-9.31	4.17	17.86
Rician GD, 100 updates	-10.29	17.40	23.50
Rician GD, stopping rule	-10.27	45.37	23.75
Amplitude GD, 10 updates	-8.80	3.29	20.21
Amplitude GD, 100 updates	-9.65	8.87	26.96
Amplitude GD, stopping rule	-9.61	20.93	27.11
Rician GD, hard Hankel	-11.06	47.39	20.93
Linearized least squares	-6.92	1.06	19.86

NMSE and severe-error rates average seven primary SNR conditions equally. Time is the median at SNR 5 dB, RSR 12 dB. Learned scores average three seeds; classical cases are not counted three times.

The EM model's average gain over HS-GS is 1.675 dB, with interval $[1.618, 1.730]$ dB and seed-specific gains of 1.533–1.787 dB (Fig. 2). At 20 dB, the gain is 0.139 dB with interval $[-0.017, 0.294]$ dB, which does not establish a clear improvement.

Severe errors fall from 14.25% for HS-GS to 0.45% for the proposed method. In the tested SNR mixture, that is roughly one badly wrong estimate in seven cases versus one in 220.

Median time is 8.24 ms versus 525.62 ms for HS-GS, about 64 times faster (Table II). Timing includes initialization and rank search, using one Apple M2 CPU thread, double precision, batch size one, 100 cases and three repeats at SNR

TABLE III
FOLLOW-UP AVERAGE GAINS (DB)

Comparison	Set 1	Set 2
New-data adaptive vs. global filtering	2.252	2.185
New-data adaptive vs. no filtering	4.250	4.191
New-data adaptive vs. HS-GS	1.939	1.837
Original adaptive vs. supervised recipe	1.015	1.012
Original adaptive vs. final-only filtering	0.697	0.694
Fixed-open mix vs. original adaptive	0.064	0.066

5 dB and RSR 12 dB. These are implementation-specific measurements: learned methods use PyTorch and classical methods use NumPy/SciPy. Offline training is separate. No filtering is faster but less accurate.

C. Operating Limits

The proposed method loses 1.617 dB at RSR 0 dB, 1.104 dB with two paths, and 0.203 dB with three paths (Fig. 3). Good average performance does not ensure good performance with a weak reference or a simple channel.

Transfer tests give gains for noise inputs wrong by ± 3 dB and three channel alternatives: four clusters of ten rays within $\pm 5^\circ$, ten equal-power paths, or ten paths with geometrically decreasing powers. Alternatives match base-channel energy and are tested only at SNR 5 dB, RSR 12 dB. Array-size and user-count changes do not all help. Path counts overlap training; these tests do not resolve failures under earlier protocols.

D. Further Comparisons

A follow-up trains 15 models and uses two fresh test sets, each with seven SNR settings and two difficult settings of 200 cases. Six primary comparisons are adjusted for multiple testing per set. Table III's 1.939 and 1.837 dB gains over HS-GS use independently generated training data; they are separate from the main 1.675 dB result. Each new training dataset uses one fixed seed, limiting replication.

The supervised comparison keeps architecture, input pilots and training budget, but trains on true channels using mean linear NMSE. The label-free model leads by about 1 dB under the main averaging rule. Selecting the supervised checkpoint by withheld likelihood reduces this to about 0.73 dB; pooling linear errors across SNRs reduces it to about 0.36 dB. Supervision wins in some settings, including two paths, and transfers less consistently from 21 to 30 input pilots in validation replay. This is not a general advantage of self-supervision over supervision.

Filtering at every stage gains about 0.70 dB over final-only filtering. A separate timing sample gives 4.65 ms versus 2.05 ms; it is not combined with main-study timing. The mixing factor is usually near one. Fixing it at one gives a small mean gain but worsens two of three seeds on two-path cases. It still stores 188 parameters. We retain the original method; the mixing factor is not a reliable on/off decision.

E. Which Inputs Help?

We compare full, spectrum-only, noise/residual-only and size/stage-only controllers. Size and stage remain in every version. All use the same update and filter, trained from scratch on 4,096 shared cases, with three seeds and 12 epochs. Two fresh sets contain 2,304 cases: seven SNR settings and two difficult settings, 128 cases per condition per set.

Full inputs gain 1.22–1.25 dB over spectrum-only, 0.57–0.58 dB over noise/residual-only, and 2.21–2.23 dB over size/stage-only inputs (Fig. 4). All three seeds improve in both sets; adjusted intervals remain positive. Final and selected checkpoints give the same scores. Yet full inputs lose 0.39–0.40 dB against noise/residual-only inputs at SNR 5 dB, failing the declared 0.25 dB protection requirement. They also lose 0.82–0.93 dB on two-path cases.

Both input groups help the average, not every channel. This smaller study does not replace the main study. All models store 188 parameters; masked weights are inactive. Removing controller inputs leaves noise variance in the measurement update and channel information in the filter.

VI. DISCUSSION AND CONCLUSION

A small model can learn useful Hankel filtering without true-channel targets. Its practical setting is repeated estimation near the simulated training population, where offline training is reused. The average accuracy and time gains come with weak-reference and few-path losses.

All results are simulations. Reference calibration is assumed known, and no physical receiver measurements are used. Three seeds sharing training cases are not three independent datasets. We have not proved unique channel recovery in every setting or convergence of the full learned estimator. The comparisons also do not establish superiority over published URformer performance. Noise-dependent regularized classical controls and broader independent training/pilot replications would be needed for a stronger robustness claim. We do not infer the cause of a loss solely from its association with SNR, path count or reference strength.

APPENDIX
INITIALIZATION AND NUMERICAL CHECKS

For element n, form $A_n \in \mathbb{C}^{P\times(K+1)}$ with rows $[s_p^T, b_{np}]$. Let u_n be the dominant eigenvector of $A_n^H \operatorname{diag}(z_n) A_n$ and set

$$c_n = \frac{\sum_p z_{np} |(A_n u_n)_p|}{\sum_p |(A_n u_n)_p|^2}. \tag{15}$$

The first K components of $c_n u_n \exp[-j \arg(u_{n,K+1})]$ initialize the channel row, using only 21 input pilots in training. Checks cover likelihood derivatives, noise scaling, finite gradients, repeated singular values, split isolation, single/double agreement and checkpoint recovery. All 432 input-study result files and 24 checkpoint replays passed the saved-output audit.

One finite-input SVD failure occurred in the classical HS-GS array-size test ($N = 64$). The same matrix was recomputed using SciPy's alternative SVD driver; the recorded relative reconstruction error was below 3×10^{-15}. The trial was

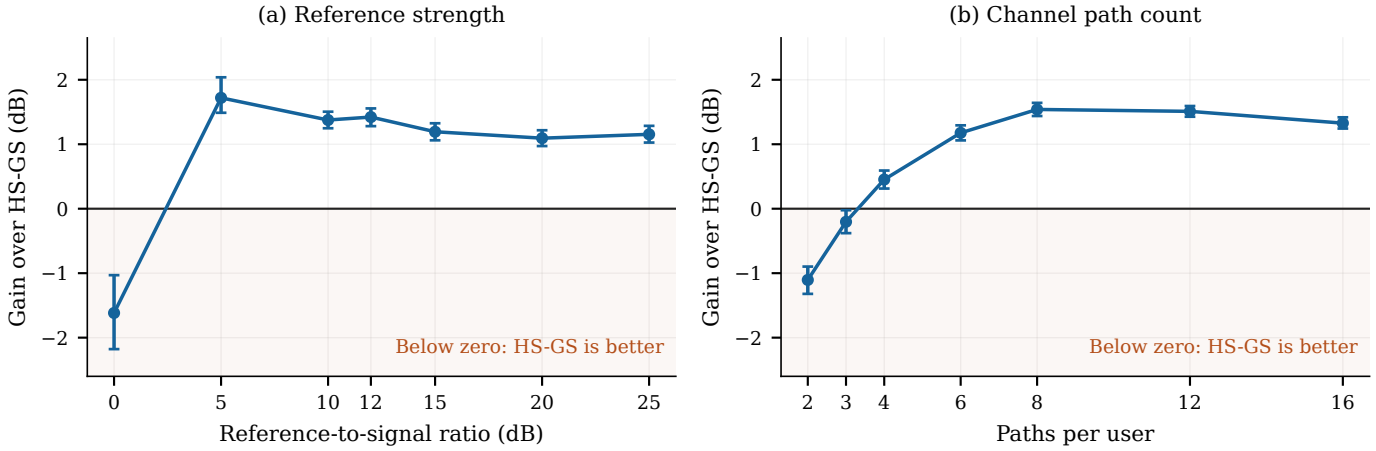


Fig. 3. Limits at SNR 5 dB: (a) reference strength with three-to-seven-path channels; (b) fixed path counts at RSR 12 dB. Positive gain favors the proposed estimator. Bars are conditional pointwise paired-case 95% intervals. Path counts overlap training.

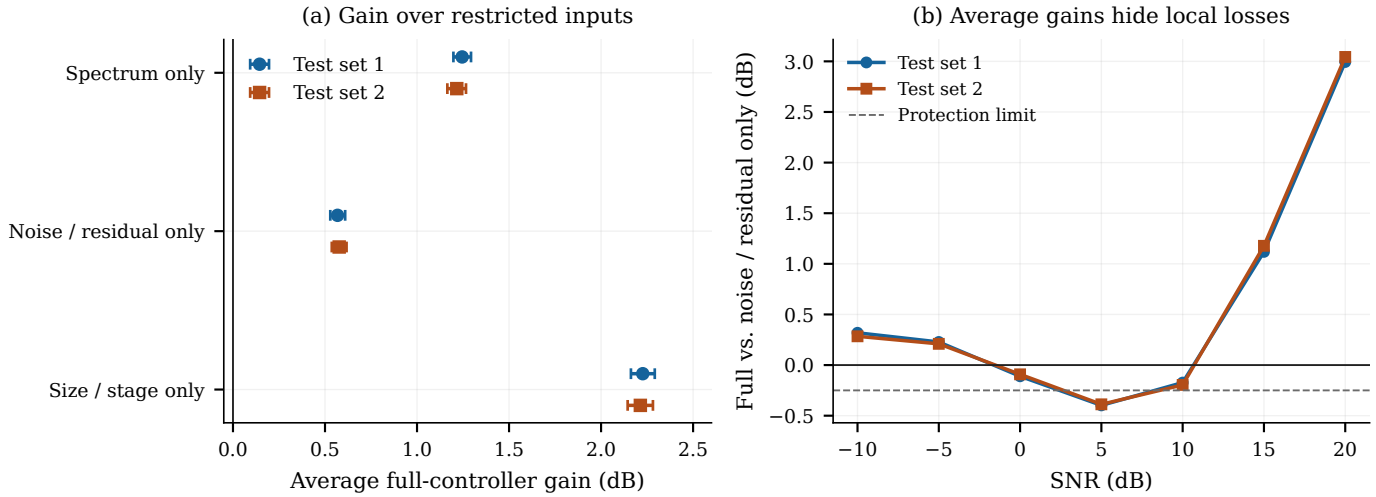


Fig. 4. Smaller-budget input study. (a) Full-controller gains over restricted inputs; intervals adjust for three comparisons per set. (b) Full versus noise/residual-only inputs, including the loss at 5 dB. The dashed line is the declared 0.25 dB protection limit. Both fresh sets have 128 cases per condition.

retained, and its inputs, equations and iteration budget were unchanged.

The reproducibility package has code, configurations, checkpoints, compact errors, timings, hashes, seeds and a replay observation subset, not full observation/prediction arrays. Software: Python 3.12, NumPy 2.4.6, SciPy 1.17.1 and PyTorch 2.13.0.

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